

AI vs ML vs DL: Classification Tasks & ML Problem Formulation

June 13, 2026

1 Part A: Classify the Following Tasks as AI, ML, or DL

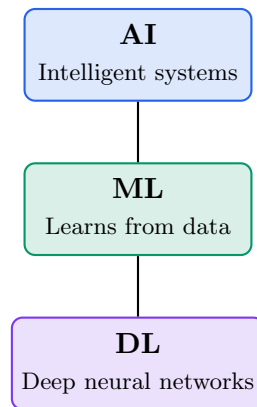
Task	Category	Reason
Playing chess using predefined rules	AI	Uses logic and decision-making.
Email spam detection	ML	Learns patterns from labeled emails.
House price prediction	ML	Predicts numerical values from historical data.
Face recognition in photos	DL	Uses deep neural networks for image processing.
Self-driving car vision system	DL	Processes camera images using deep learning.
Voice assistants (Siri, Alexa)	AI + ML + DL	AI application powered by ML and DL models.
Product recommendation on shopping websites	ML	Learns user preferences from data.
Language translation (English → Hindi)	DL	Uses deep neural networks such as transformers.
Expert medical diagnosis system with rules	AI	Rule-based decision-making system.
Handwritten digit recognition	DL	Uses neural networks to identify images.

Table 1: Classification of tasks into AI, ML, and DL.

1.1 Quick Difference

- **AI (Artificial Intelligence):** Makes machines behave intelligently.
- **ML (Machine Learning):** Subset of AI that learns from data.
- **DL (Deep Learning):** Subset of ML that uses multi-layer neural networks.

1.2 Hierarchy



2 Part B: ML Problem Statement with Features and Labels

2.1 Problem Statement

Predict whether a customer will buy a product after visiting an online store.

2.2 Dataset

Age	Income	Time on Website (min)	Purchased?
25	30000	5	No
40	70000	15	Yes
30	50000	8	No
45	90000	20	Yes

Table 2: Sample customer dataset.

2.3 Features (Input Variables)

The information used to make predictions:

1. Age
2. Income
3. Time on Website

Feature vector:

$$X = [\text{Age}, \text{Income}, \text{Time on Website}]$$

2.4 Label (Target Variable)

The value we want to predict:

$$Y = \text{Purchased?} \quad \text{where} \quad \text{Yes} = 1, \quad \text{No} = 0$$

2.5 ML Objective

Learn a function

$$Y = f(X) \iff \text{Purchased} = f(\text{Age}, \text{Income}, \text{Time on Website})$$

so that for a *new* customer, the model can predict whether they are likely to purchase the product.

2.6 Type of Problem

Since the label has two categories (**Yes** or **No**), this is a

Binary Classification Problem (Machine Learning)

2.7 Another Example: House Price Prediction

Features:

$$X = [\text{Area}, \text{Bedrooms}, \text{Age_of_House}]$$

Label:

$$Y = \text{House_Price}$$

Since the label is a numeric value, this is a

Regression Problem (Machine Learning)

Summary

Term	Meaning
Feature	Input variable used for prediction
Label	Output/target variable to be predicted
Classification	Predicting categories (Yes/No, Spam/Not Spam)
Regression	Predicting numeric values (Price, Salary)
AI	Broad field of intelligent systems
ML	AI that learns from data
DL	ML using deep neural networks

Table 3: Key terminology summary.